

Spacecraft Attitude Control Simulator (SACS)

By Bricen Rigby



CAL POLY

Project Overview

Objective:

Develop a single-axis attitude control simulator to replicate the frictionless environment of space and test cold gas thruster control schemes.

Key Goals:

- Validate control logic via real-time testing
- Compare observed vs. simulated dynamics
- Improve fidelity and efficiency of spacecraft simulator testbeds

Hardware & Assembly

Key Components:

- **Air Spindle:** Provides near-frictionless rotation using porous carbon
- **Cold Gas Thrusters:** Controlled via solenoid valves & moment arms
- **Frame:** Constructed from aluminum extrusion for modularity
- **Paintball Bottle:** Supplies compressed air

Fusion 360 modeling:

- Used to determine mass properties & optimize nozzle arm placement

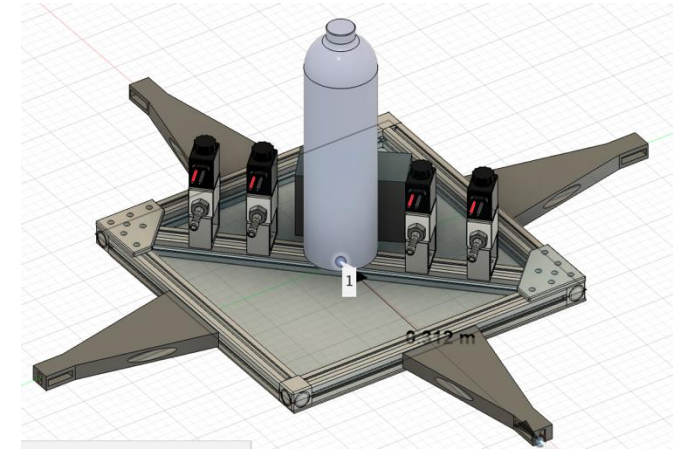


Figure 1. SACS Platform as modeled in Fusion 360

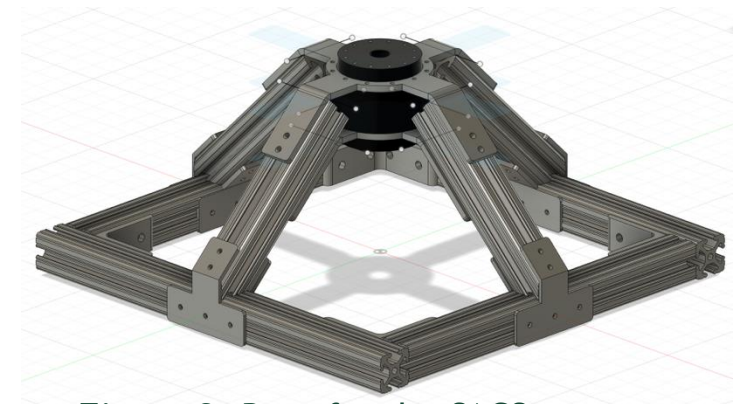


Figure 2. Base for the SACS Platform, containing the air spindle

Control System & Simulation

Hardware Logic:

- Teensy microcontroller + MOSFETs to actuate solenoids
- Diodes for back-EMF protection
- Custom PCB for robustness over breadboarding

Simulink Control Model:

- Integrates equations of motion
- PD control (K_p/K_d gains) + Schmitt Trigger logic
- Simulates torque command and valve switching behavior

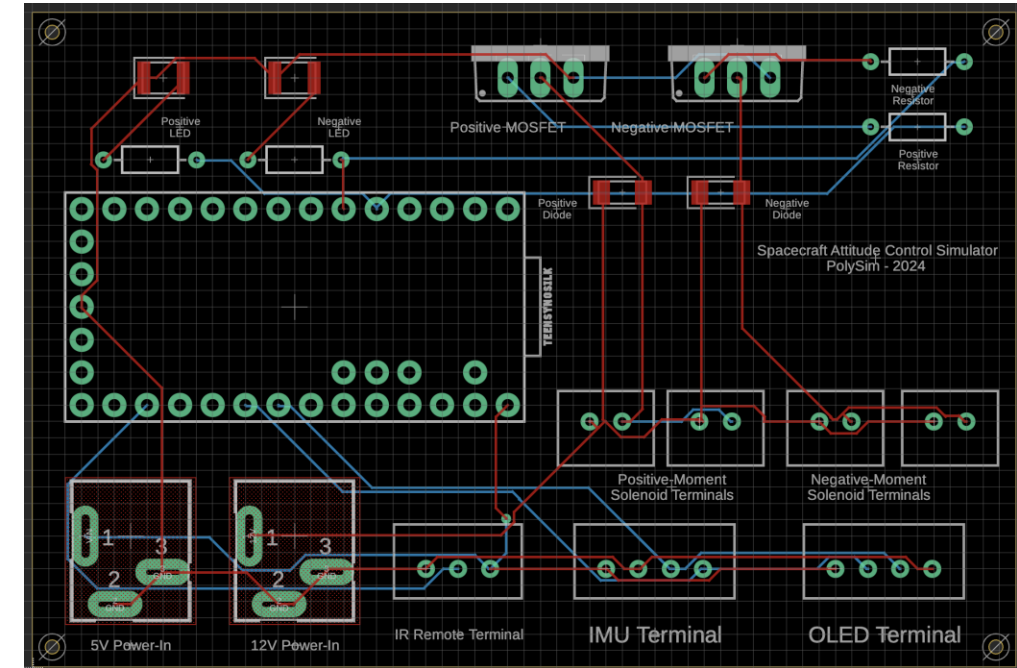


Figure 3. Custom Printed Circuit Board Design in Fusion 360

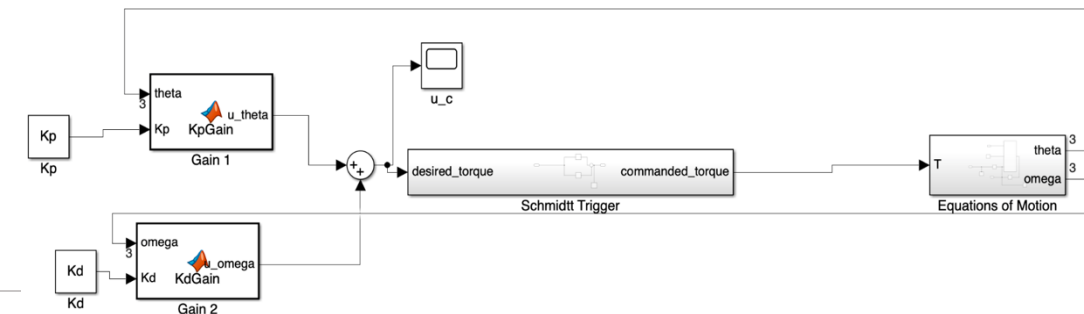


Figure 4. Simulink Model of the SACS Platform

Results & Validation

Simulated Data

Settling Time Comparison:

- Simulated: 14 seconds
- Observed: 13 seconds
- Error: ~7% (likely from underestimating actual thrust)

Implication:

Platform effectively replicates low-friction rotational dynamics and allows for real-world control scheme testing.

Observed Data

